PRE-ALGEBRA (7TH) · UNIT 8: LINEAR FUNCTIONS AND GRAPHING

Bike Share Showdown

Graph linear functions to choose the best bike or scooter rental.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your city just launched three ways to get around downtown: a bike share, an electric scooter company, and a monthly pass. Each has a different starting fee and price per minute.

You will graph each option as a linear function, compare them, and write a guide that helps riders choose.



CityBike
$1 to unlock
$0.20 per minute

ZoomScoot
$2 to unlock
$0.15 per minute

Learning Goals

- I can identify functions and represent them with tables, rules, and graphs.
- I can interpret slope and intercepts and fit a trend line.
- I can solve systems by graphing and graph linear inequalities.

Project Timeline

When	What you will do
Day 1	Functions & Data
Day 2	Three Options
Day 3	Budget
Final	Rental Comparison Guide + reflection

Part 1: Functions & Data

Day 1 · Use the Ride Data card.

Ride Data: minutes vs. cost for 8 CityBike rides

Ride 1	Ride 2	Ride 3	Ride 4
5 min · $2.00	10 min · $3.00	12 min · $3.40	15 min · $4.00
Ride 5	**Ride 6**	**Ride 7**	**Ride 8**
20 min · $5.00	25 min · $6.00	30 min · $7.00	8 min · $2.60

1. Is the set of (minutes, cost) pairs a function? Could a different set of rides make it not a function? Explain.

2. Make a scatter plot of the ride data. Describe the correlation.

3. Find the slope using two rides and write the equation of the line.

4. Predict the cost of a 45-minute ride.

Part 2: Three Options

Day 2 · ZoomScoot: $2 + $0.15 per minute. MonthPass: $30 per month, rides free.

5. Write rules for ZoomScoot and MonthPass for x total minutes in a month.

6. Graph CityBike, ZoomScoot, and MonthPass (for one long ride per month) on the same axes. Where do CityBike and ZoomScoot cross?

7. A rider takes 20 rides of 15 minutes each in a month. Find the monthly cost of each option.

8. Solve the system y = 0.20x + 1 and y = 0.15x + 2 algebraically to check your graph.

Part 3: Budget

Day 3 · You have at most $12 per week for CityBike rides.

9. Let r be the number of rides and m the minutes per ride (same each time). Write the cost of r 10-minute rides and an inequality for the budget.

10. Graph $y \le 0.20x + 1$ for one ride. Which side is shaded and what does a point in the shaded region mean?

11. Write your Rider Guide: which option is best for (a) one short trip, (b) one long trip, (c) daily commuting? Use your graphs.

Final Product: Rental Comparison Guide

Your final product must include:

- Scatter plot with trend line
- Rules and graphs for three options
- System solved by graphing and algebra
- Budget inequality
- Rider Guide with recommendations

Reflection

R1. How do slope and y-intercept help you compare plans without graphing?

R2. Why is the intersection of two lines important here?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Linear functions, slopes, intercepts, systems, and inequalities are modeled and used correctly to compare options. Goes beyond the minimum with a creative, realistic choice.	Linear functions, slopes, intercepts, systems, and inequalities are modeled and used correctly to compare options.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Rider Guide includes graphs, rules, intersection points, and a clear decision chart. Polished, organized, and visually engaging.	Rider Guide includes graphs, rules, intersection points, and a clear decision chart. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.